

How It Works

The card suit symbols don't exist on your keyboard, which is why we call the `chr()` function to create them. The integer passed to `chr()` is called a Unicode *code point*, a unique number that identifies a character according to the Unicode standard. Unicode is often misunderstood. However, Ned Batchelder's 2012 PyCon US talk "Pragmatic Unicode, or How Do I Stop the Pain?" is an excellent introduction to Unicode, and you can find it at <https://youtu.be/sgHbC6udIqc/>. Appendix B gives a full list of Unicode characters you can use in your Python programs.

```
1. """Blackjack, by Al Sweigart al@inventwithpython.com
2. The classic card game also known as 21. (This version doesn't have
3. splitting or insurance.)
4. More info at: https://en.wikipedia.org/wiki/Blackjack
5. View this code at https://nostarch.com/big-book-small-python-projects
6. Tags: large, game, card game"""
7.
8. import random, sys
9.
10. # Set up the constants:
11. HEARTS   = chr(9829) # Character 9829 is '♥'.
12. DIAMONDS = chr(9830) # Character 9830 is '♦'.
13. SPADES   = chr(9824) # Character 9824 is '♠'.
14. CLUBS    = chr(9827) # Character 9827 is '♣'.
15. # (A list of chr codes is at https://inventwithpython.com/charactermap)
16. BACKSIDE = 'backside'
17.
18.
19. def main():
20.     print('''Blackjack, by Al Sweigart al@inventwithpython.com
21.
22.     Rules:
23.         Try to get as close to 21 without going over.
24.         Kings, Queens, and Jacks are worth 10 points.
25.         Aces are worth 1 or 11 points.
26.         Cards 2 through 10 are worth their face value.
27.         (H)it to take another card.
28.         (S)tand to stop taking cards.
29.         On your first play, you can (D)ouble down to increase your bet
30.         but must hit exactly one more time before standing.
31.         In case of a tie, the bet is returned to the player.
32.         The dealer stops hitting at 17.''' )
33.
34.     money = 5000
35.     while True: # Main game loop.
36.         # Check if the player has run out of money:
37.         if money <= 0:
38.             print("You're broke!")
39.             print("Good thing you weren't playing with real money.")
40.             print('Thanks for playing!')
41.             sys.exit()
42.
43.         # Let the player enter their bet for this round:
```